

How Certain is the AI?

Conformal Inference for Classification Models

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GOAL: To output efficient, adaptive, and usable prediction sets

What is Conformal Inference?

Neural networks are **miscalibrated** even when they are accurate; wrong classification is disguised with high confidence. We want to accurately flag uncertainty.

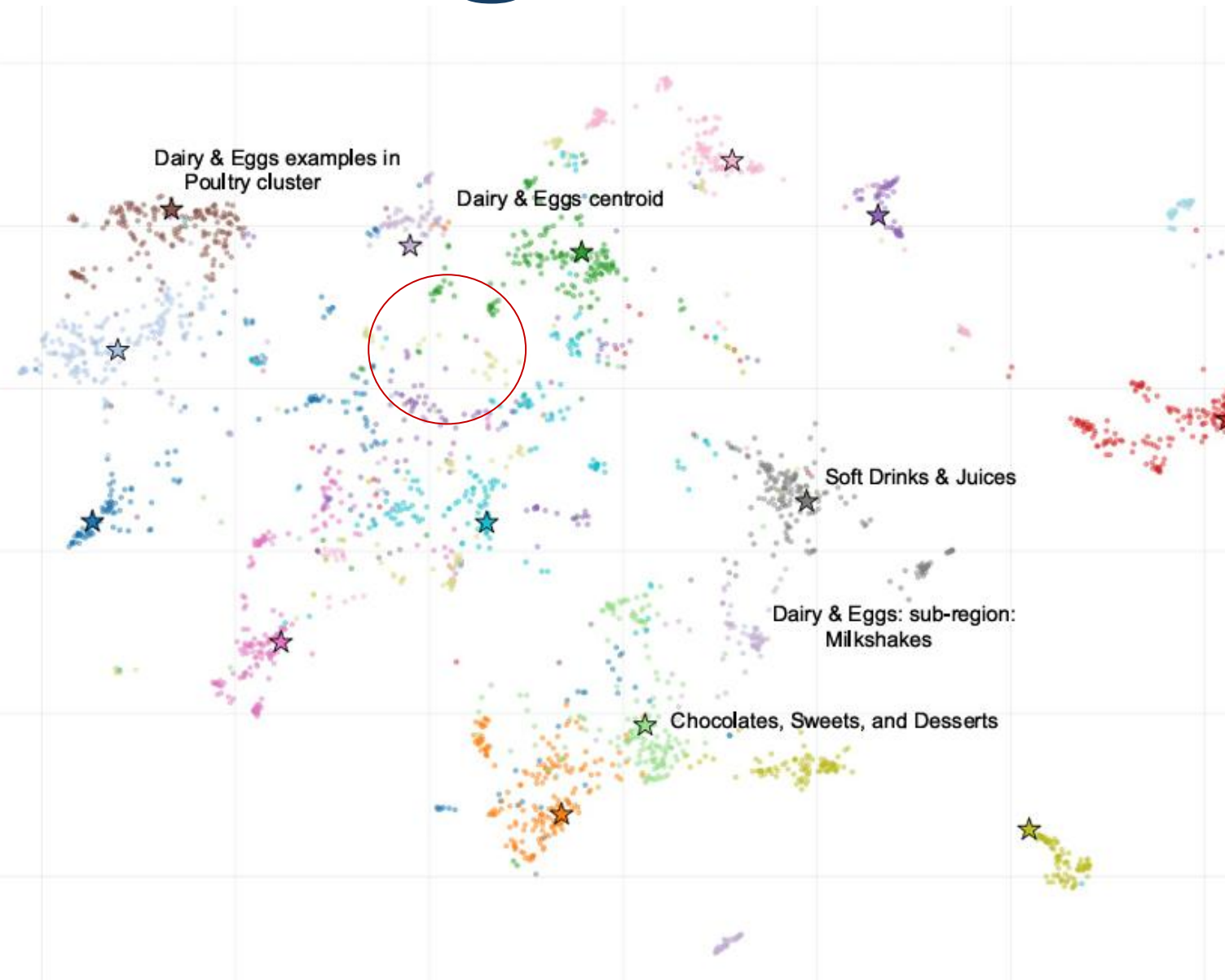
Conformal Inference is a second layer fitted after a model. On held-out calibration data it measures how strange each point looks given a candidate label, then decides a threshold, returning the plausible labels as a **prediction set**, guaranteed to hold the true label 95% of the time. Distribution-free, no retraining, with any model.

Design Problem: Reading uncertainty from the embedding space

The 95% guarantee holds for **any** measure of strangeness (non-conformity score). A poor score reaches 95% by returning most candidate labels every time.

We used the embedding space to develop a score, exploring which geometry carries information on uncertainty.

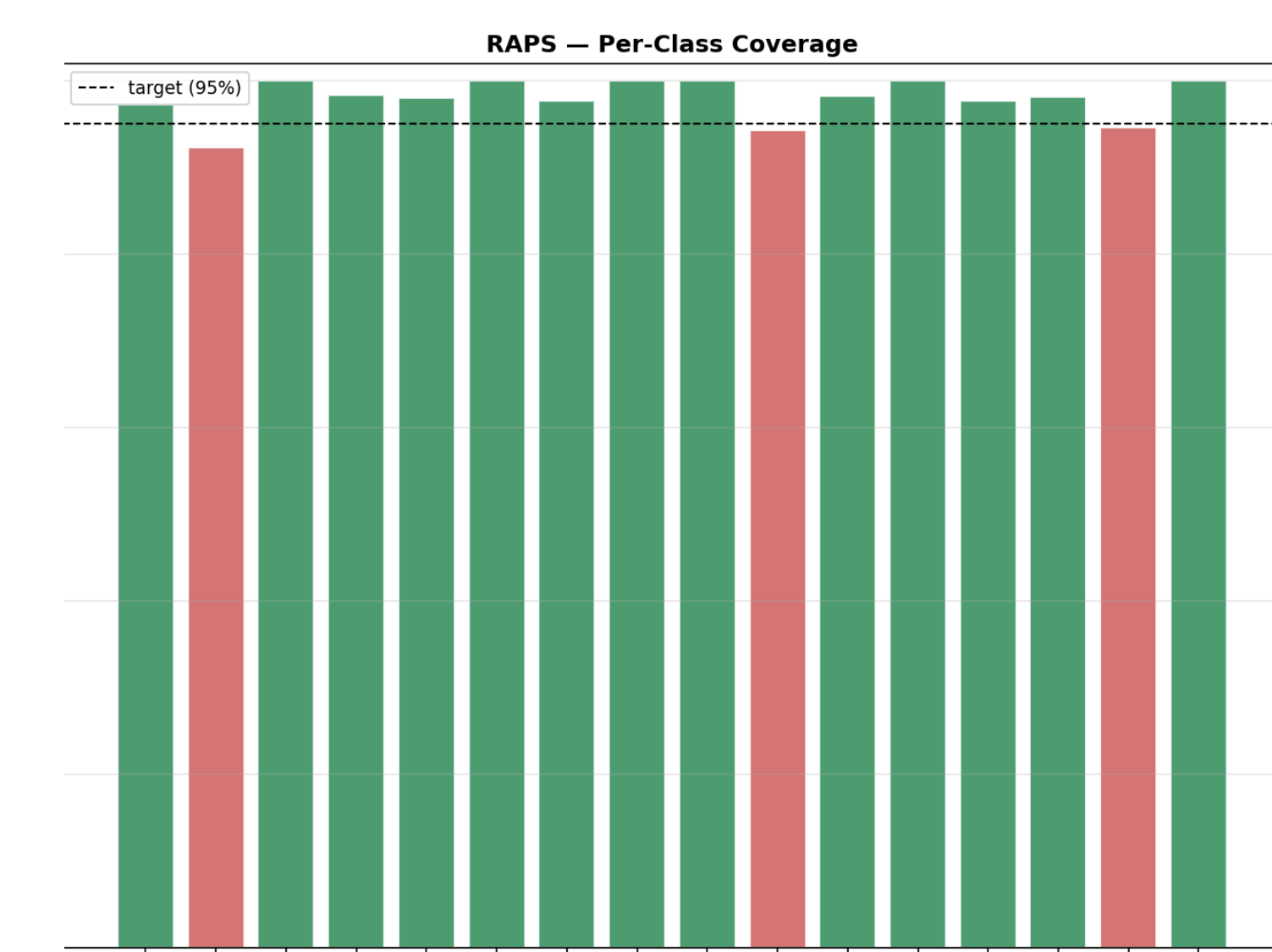
Findings:



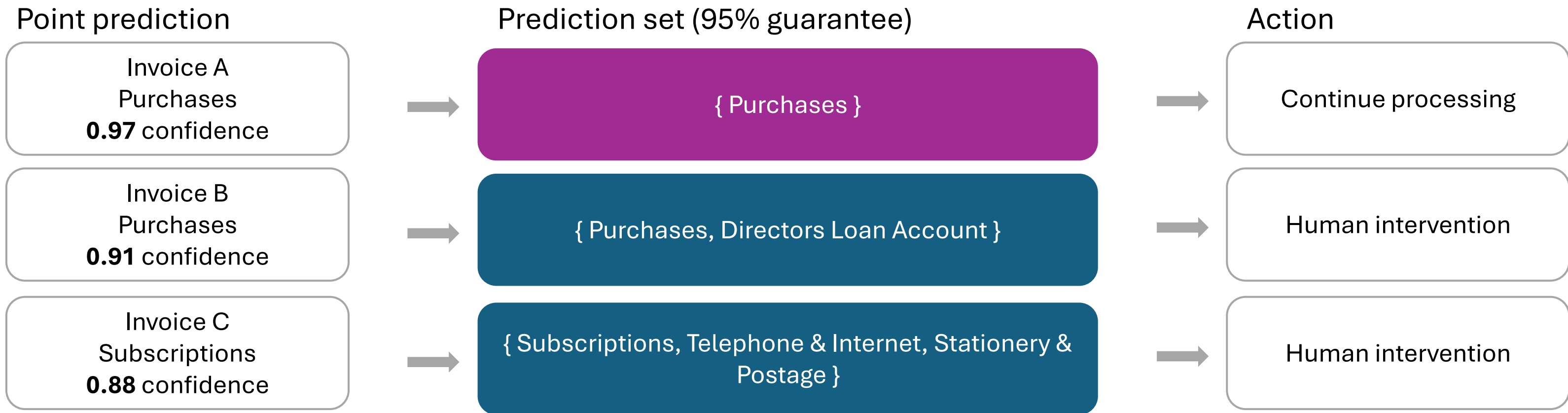
AMuRD embedding space. Multi-modal class has no meaningful centroid, so centroid score measures the wrong thing.

Caveat: Coverage is marginal

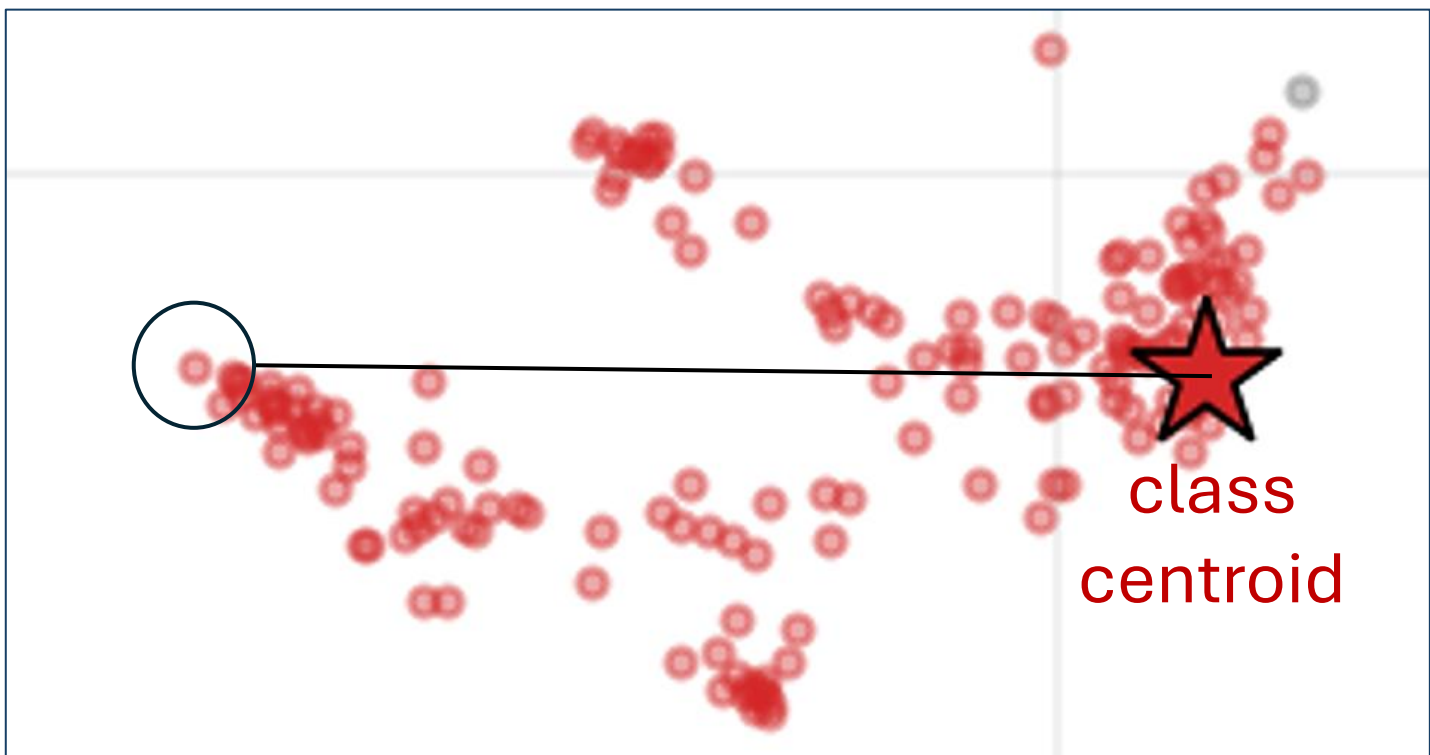
The 95% guarantee is **marginal**: it holds across the test set as a whole. Harder categories are undercovered and easier ones overcovered, and average coverage (97.6% on AMuRD) conceals the categories below target.



Per-class coverage on AMuRD; 3 categories don't meet the coverage guarantee; Beef & Processed Meat, Rice, Pasta & Pulses, Vegetables & Fruits



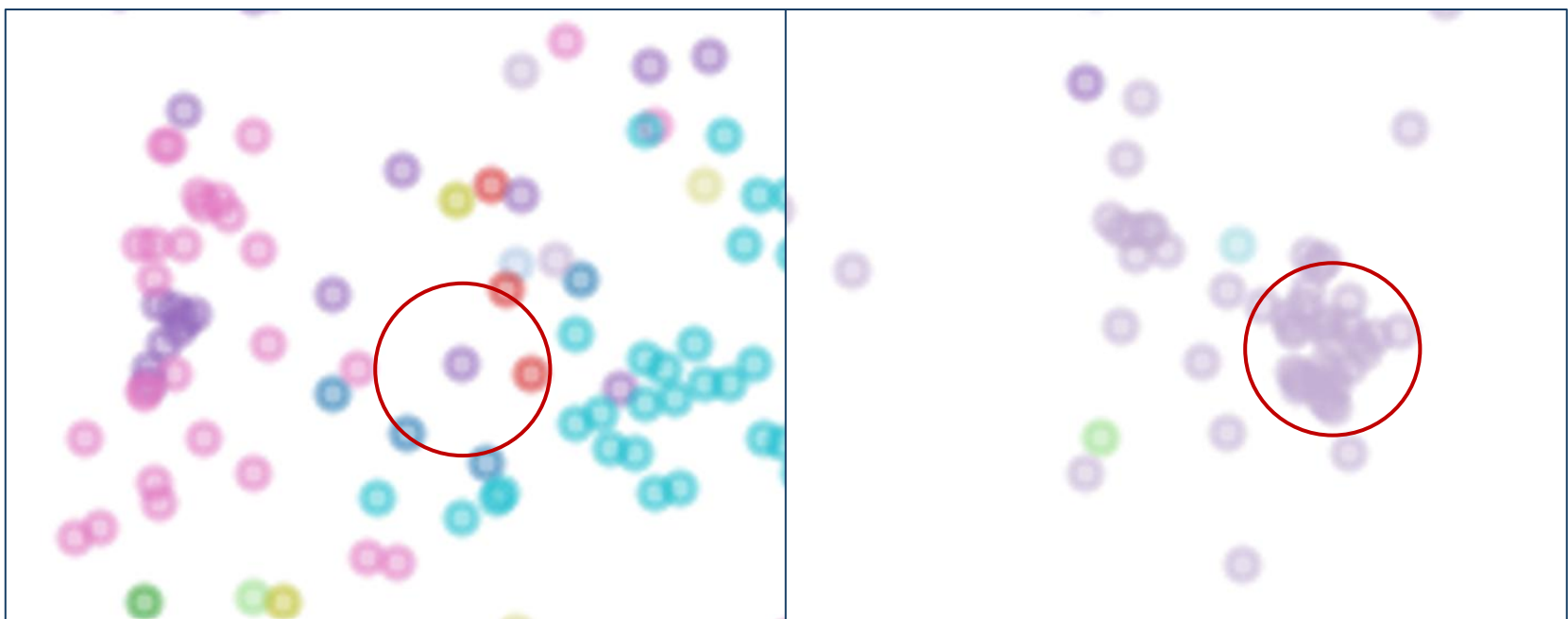
Global Geometry: Typicality



Is this document a representative example of the candidate label?

We measured distance to the class centroid under Euclidean and cosine metrics, testing if direction was an uncertainty indicator outside of magnitude, then rescaled the distance with Mahalanobis distance which considers class variance in different directions.

Local Geometry: Nearest Neighbours



High uncertainty
The neighbours are a mixture of categories

Low uncertainty
The neighbours carry the same label

How many of the nearby points belong to the candidate class?

The non-conformity score is the fraction of the k nearest embeddings that **disagree** with the candidate class. There are no assumptions about the shape of a class.

This reflects **aleatoric uncertainty**: irreducible uncertainty in the data

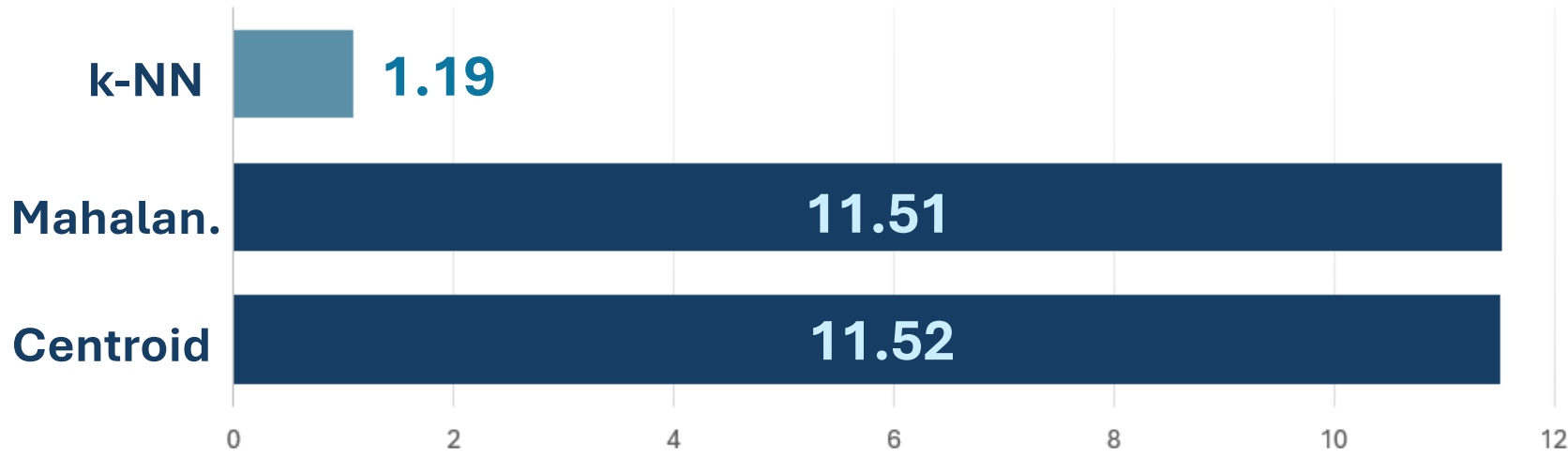
1. The problem with centroid distance:

Real categories are not single clusters. Often they are **multi-modal**, like “Dairy & Eggs” with subclusters for eggs, milkshakes, etc. This means the centroid ends up in a low-density gap between them, far from many actual members of the class.

Therefore, the standard distance-to-centroid is inflated, and prediction sets grow. Mahalanobis distance doesn't help this problem as it comes from reducing the classes to a single point.

2. Local geometry gives better results:

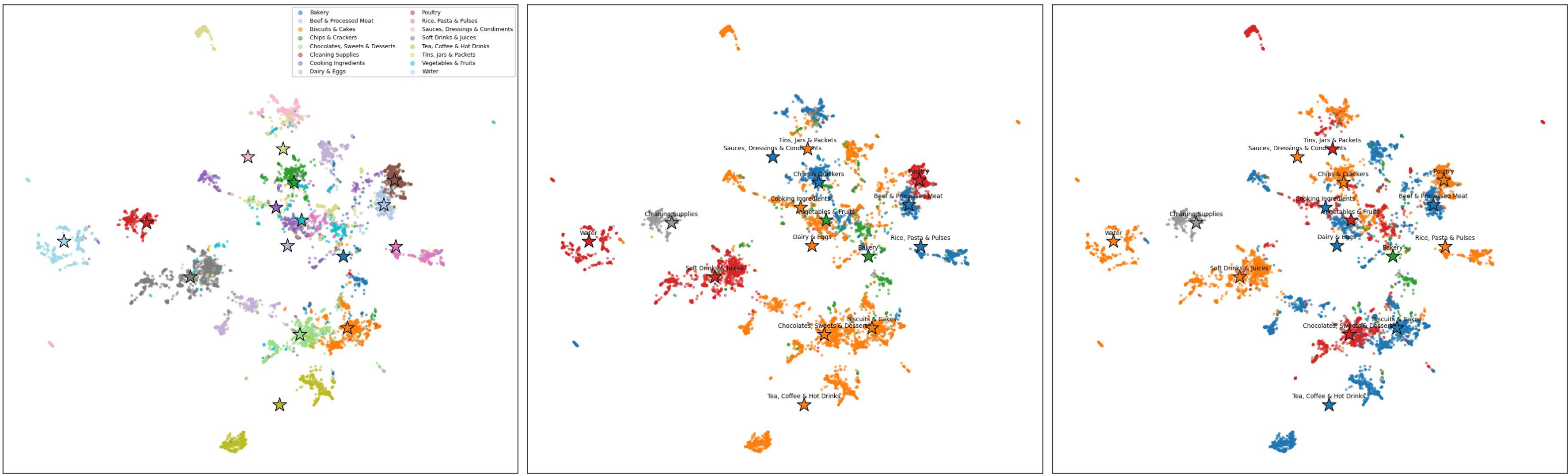
Summarising classes into single points costs the set size. Reading the neighbourhood produces sets small enough to act on.



Average set size on AMuRD across embedding-based scores, out of 16 categories.

Next Steps: Clustering the embedding space

Calibrate conformal inference within groups of similarly behaving classes, so the guarantee holds in every group rather than just on average. We grouped by score distribution, reflecting real structure but affected by limited data (<40 calibration examples per group). Next steps should explore further partitioning methods.



Clusters on AMuRD using two different score distributions; LAC (middle, softmax-based) and k-NN (right) compared to class groupings.

Research completed...

...on **3 datasets**: FashionMNIST, AMuRD, and 2,169 invoices over 12 categories

...across **11 non-conformity scores** from three points in the pipeline; softmax, logit, and embedding

...using **8 evaluation metrics** focusing on usability and conditional coverage

Key Takeaways

Any score gives you the 95% coverage guarantee but to build adaptive sets, small enough to act on, we need the right score. The embedding space must be read locally rather than summarising away details.

The open problem is conditional coverage: the guarantee should hold for every category and every difficult subset, not just on average. Calibrating within a fixed partition provides this (given enough data). Next steps should explore how to partition.